

TM Forum Component

Location Management

TMFC014

Maturity Level: General availability (GA)	Team Approved Date: 11-Mar-2025
Release Status: Pre-production	Approval Status: TM Forum Approved
Version 1.2.0	IPR Mode: RAND

Notice

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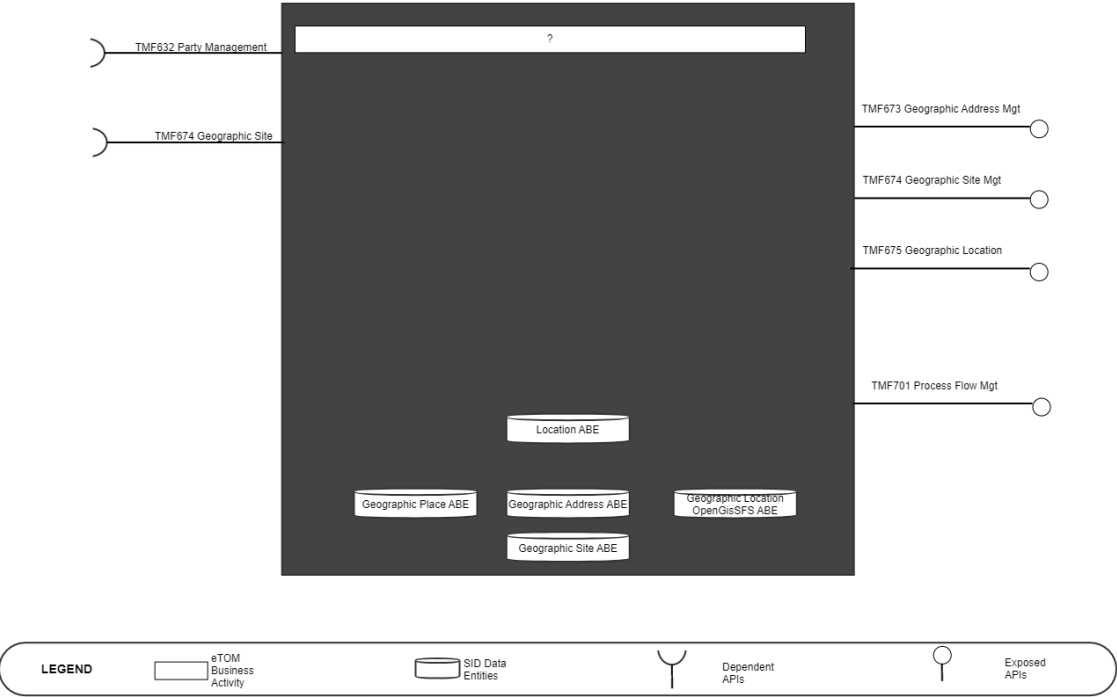
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1. Overview

Component Name	ID	Description	ODA Function Block
Location Management	TMFC014	The Location Management Component allows easy reference to geographic places important to other entities, where a geographic place is an entity that can answer the question “where?” . This component could be a facade tool into GIS systems (e.g. Google Maps) Also covers the operations to manage (create, read, delete) geographic sites that can be associated with a customer, account, service delivery or other entities. And finally gives the capabilities to retrieve /list /validate addresses that are named as structured textual ways of describing how to find a Property in an urban area (country properties are often defined differently). It allows looking data for worldwide addresses through popular GIS systems like Google Maps or government master addresses systems. It can also be used to validate geographic data, to be sure that it corresponds to a real geographic address. Finally, it can be used to look for an address by: searching an area as a start (city, town), then zooming on the streets of this area, and finally listing all the street segments (numbers) in a street.	Production Domain

TMFC014 Location Management



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:.

Identifier	Level	Business Activity Name	Description
none			

Note: as no eTOM business activity is currently responsible for Location Management, refer to JIRA paragraph.

2.2. SID ABEs

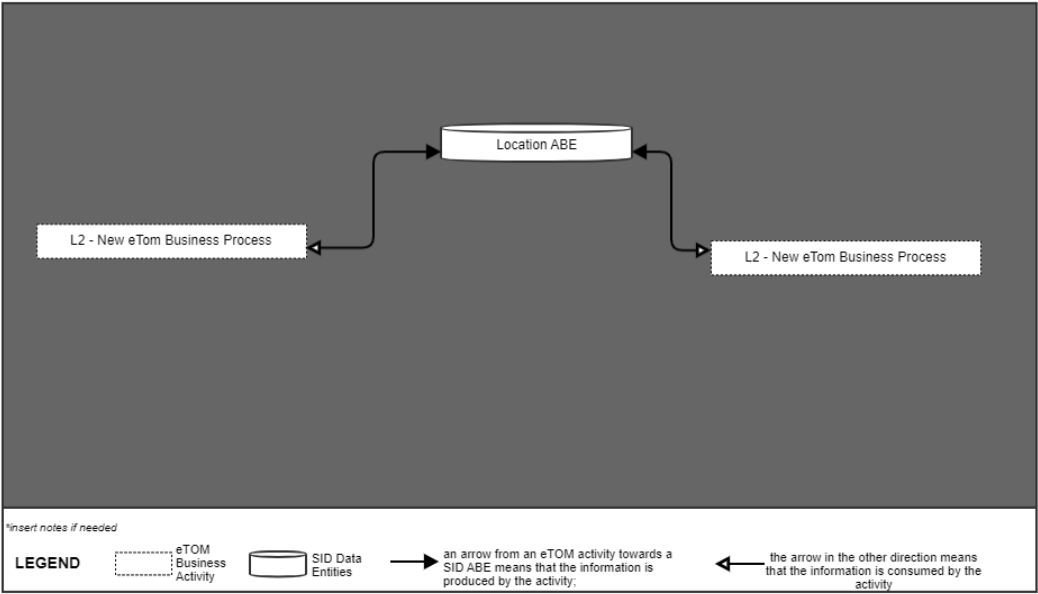
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Location ABE	Geographic Place ABE
	Geographic Location ABE
	Geographic Address ABE
	Geographic Site ABE
	Local Place ABE

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function	Description	ID	Domain	Aggregate Function Level 1	Aggregate Function Level 2
Location Change History Management	Location Change History Management; Tracks all changes of location data, making available attributes according their historical values in certain periods.	429	Operations Readiness & Support	Resource Management	Location Management
Pre-formatted Location Information Presentation	Pre-formatted Location Information Presentation generates different views for different business cases (e.g. different format of address strings)	430	Operations Readiness & Support	Resource Management	Location Management
Location Information Updating	Location Information Updating provides means to update the repository with new/updated location information from external sources.	431	Operations Readiness & Support	Resource Management	Location Management
Location Information Searching	Location Information Searching provide the ability to search for a provided location/address, as part of the Location Management, including the ability to return near matches if an exact match is not found.	432	Operations Readiness & Support	Resource Management	Location Management
Location Structure Data Configuration	Location Structure Data Configuration provides facilities for creating, modifying, and deleting location structures data according to business rules of Service Providers or national and international location regulations. Also, utilities for defining sets of location attributes, levels and hierarchies should be available.	433	Operations Readiness & Support	Resource Management	Location Management
Location Data Integrity Management	Location Data Integrity Management provides ability to maintain data integrity in the whole location repository. It's especially important if there are many external data sources that deliver new addresses for the repository.	434	Operations Readiness & Support	Resource Management	Location Management

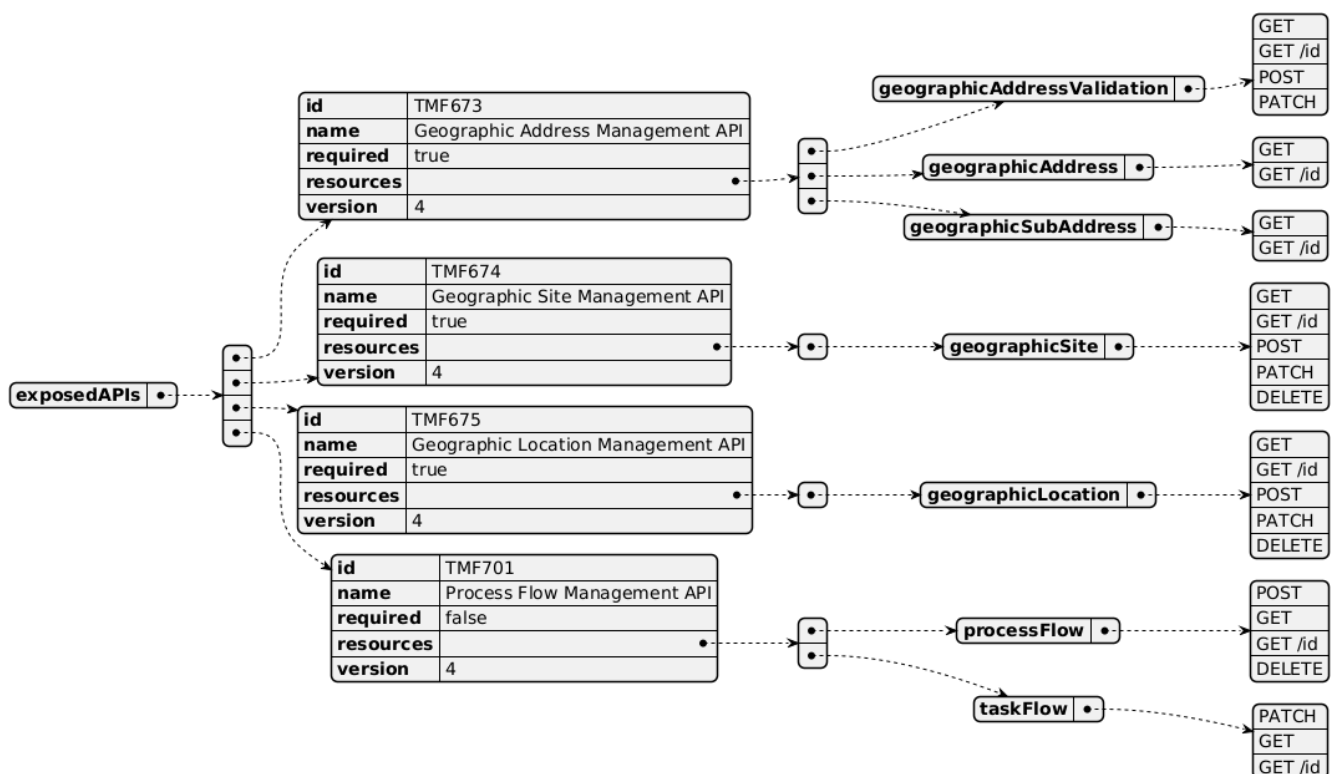
3. TM Forum Open APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component. At this stage we list the APIs, resource and operation we no mention to optionality (in other word no mention about mandatory VS optional resource)
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs. At this stage **no optionality is defined** and none of this 'required' API is listed as 'mandatory'
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:

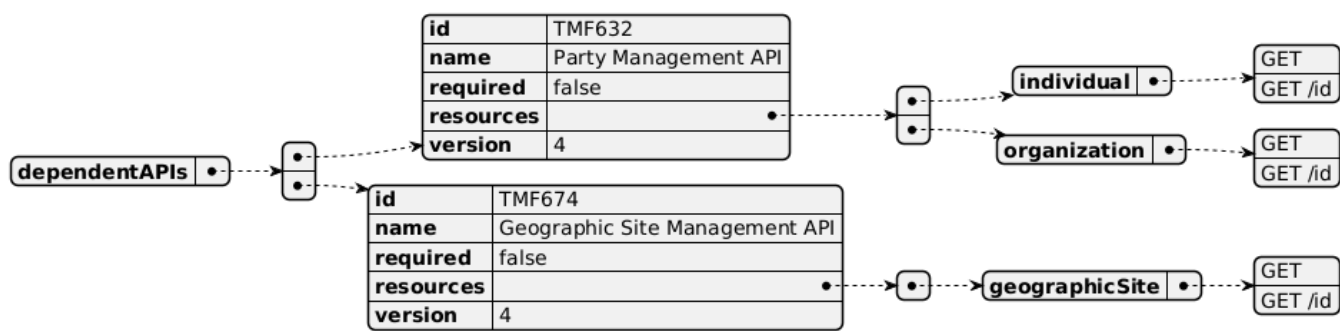


API ID	API Name	API Version	Mandatory / Optional	Operation
TMF673	TMF 673 Geographic Address Management API	4	Mandatory	geographicAddressValidation • GET • GET /ID • POST • PATCH
TMF673	TMF 673 Geographic Address Management API	4	Mandatory	geographicAddress • GET • GET /ID
TMF673	TMF 673 Geographic Address Management API	4	Mandatory	geographicSubAddress • GET • GET /ID
TMF674	TMF 674 Geographic Site Management API	4	Mandatory	geographicSite • GET • GET /ID • POST • PATCH • DELETE
TMF675	TMF675 Geographic Location	4	Mandatory	geographicLocation • GET • GET /ID • POST • PATCH • DELETE
TMF688	TMF688 Event	4	Optional	listener • POST
TMF688	TMF688 Event	4	Optional	hub • POST • DELETE
TMF701	TMF701 Process Flow Management	4	Optional	processFlow • GET • GET /ID • POST

API ID	API Name	API Version	Mandatory / Optional	Operation
				DELETE
TMF701	TMF701 Process Flow Management	4	Optional	taskFlow - GET /ID - PATCH

3.2. Dependent APIs

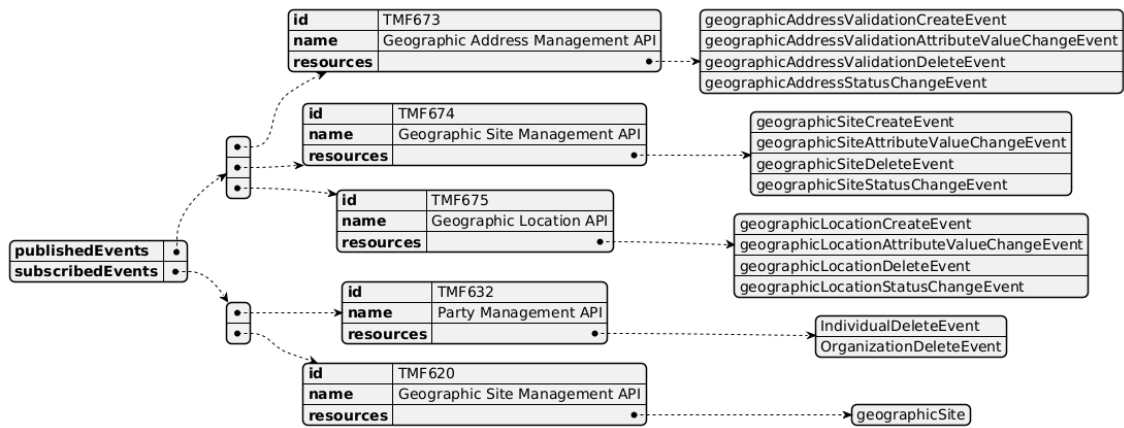
Following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	resource	Operations
TMF632	TMF632 Party Management	4	Optional		Individual / organization - GET - GET/id
TMF674	TMF674 Geographic Site	4	Optional		geographicSite - GET - GET/id
TMF688	TMF688 Event	4	Optional		event - GET - GET/id

3.3. Events

The following diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component Map on the TM Forum website for the machine-readable component specification files for this component. [TM Forum - ODA Component Directory](#).

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	23.5
eTOM	23.5
Functional Framework	23.5

5.2. eTOM

[ISA-559](#) Enrich eTOM with business activities necessary to manage any type of Location

5.3. Further Resources

n/a

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1.0	28 Jun 2022	Gaetano Biancardi , Sylvie Demarest	Final edits prior to publication
0. 2.0	19 Aug 2022	Anastasios Sarantis	Changes asked from the community . Component description changed according to feedback + Functional framework updates
1.0.0	07 Dec 2022	Goutham Babu	Initial release of document
1.0.1	25 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository
1.0.1	15 Aug 2023	Amaia White	Final edits prior to publication
1.1.0	13 May 2024	Hugo Vaughan / Gaetano Biancardi	Align to Frameworks 23.5 Align to latest template Removed TMF688 from exposed and dependent API
1.1.0	12 Jul 2024	Amaia White	Final edits prior to publication
1.2.0	24 Oct 2024	Gaetano Biancardi	Exposed API: fixed duplication of TMF674
1.2.0	14 Mar 2025	Rosie Wilson	Updated Maturity Level to Beta as confirmed by Gaetano Biancardi as v4 of TMF675 is still in pre view and API confirmed as Mandatory for this component by Anastasios Sarantis . Final administrative updates

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	07-Dec-23	Goutham Babu	Initial release of document
Pre-production	23-Jan-23	Adrienne Walcott	Updated to member evaluation status
Pre-production	15-Aug-2023	Amaia White	New release 1.0.1

Release Status	Date Modified	Modified by:	Description of changes
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14 Mar 2025	Rosie Wilson	New release 1.2.0
Production	09-May-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.1.3. Acknowledgments

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Sylvie Demarest	Orange	Reviewer
Ian Turkington	TM Forum	Additional Input
Hugo Vaughan (TM Forum)	TM Forum	Additional Input

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer